

Creating instance automatically on aws console using python boto3 from a local machine .

Requirements

Local machine +vscode/pycharm IDE

Python latest version +pip +boto3

Aws secrete storage manager(ssm)

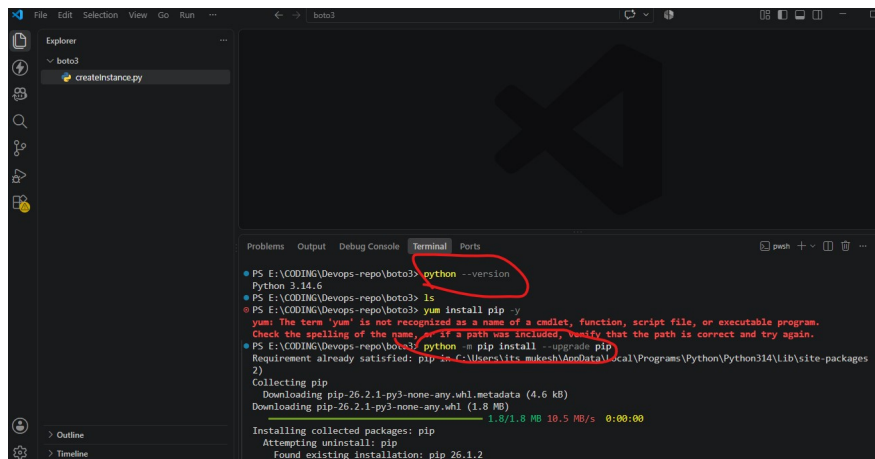
Create a directory boto3 and Open folder with vscode and install python +pip +boto3

In powershell :

```
>winget install --id Python.Python.3 -e
```

```
>python -m pip install --upgrade pip
```

```
>pip install boto3 -y
```



```
PS E:\CODING\Devops-repo\boto3> python --version
Python 3.14.6
PS E:\CODING\Devops-repo\boto3> ls
PS E:\CODING\Devops-repo\boto3> yum install pip -y
yum: The term 'yum' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
PS E:\CODING\Devops-repo\boto3> python -m pip install --upgrade pip
Requirement already satisfied: pip in C:\Users\its_makesh\AppData\Local\Programs\Python\Python314\Lib\site-packages
(2)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
  Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
    1.8/1.8 MB 10.5 MB/s 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 26.1.2
```

Create a file **createInstance.py** and include the code .

```

import boto3

ec2 = boto3.resource(
    'ec2',
    region_name='us-east-1'
)

instances = ec2.create_instances(
    ImageId='ami-0bdc7d025135d7b49', # Replace with your AMI ID
    MinCount=1,
    MaxCount=1,
    InstanceType='t3.micro',
    KeyName='KeyNv', # Replace with your Key Pair name

    NetworkInterfaces=[
        {
            "DeviceIndex": 0,
            "SubnetId": "subnet-0af332bff2206ab95",
            "AssociatePublicIpAddress": True,
            "Groups": [
                "sg-0c387840f4e73751c"
            ]
        }
    ],

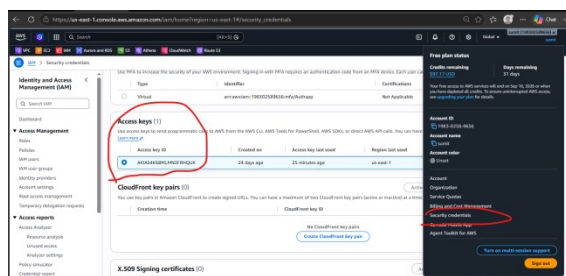
    TagSpecifications=[
        {
            'ResourceType': 'instance',
            'Tags': [
                {
                    'Key': 'Name',
                    'Value': 'Python-Created-EC2'
                }
            ]
        }
    ]
)

print("Instance created successfully!")
print("Instance ID:", instances[0].id)

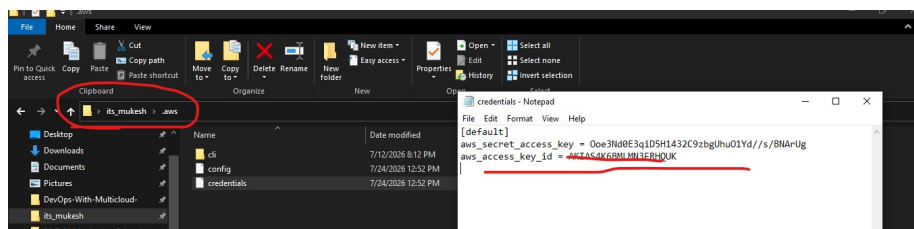
```

then configure your local machine for **SSM connect (aws configure)**.

Get the access key and secret access key from aws console



Or if you had already configured the credentials earlier you can find it on your local machine



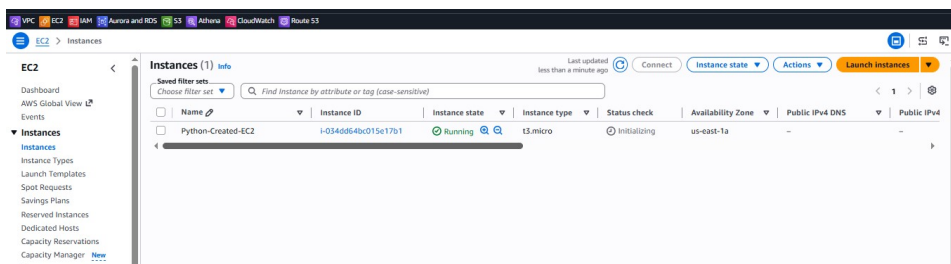
Now configure it on your local machine

```
Problems Output Debug Console Terminal Ports
PS E:\CODING\Devops-repo\boto3> python -u "e:\CODING\Devops-repo\boto3\createInstance.py
PS E:\CODING\Devops-repo\boto3> aws configure
AWS Access Key ID [*****HQUK]: AKIAS4K6BMLMN3FRHQUK
AWS Secret Access Key [*****ArUg]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json
```

Now run the file **createInstance.py**

```
Problems Output Debug Console Terminal Ports
PS E:\CODING\Devops-repo\boto3> python .\createInstance.py
Instance created successfully!
Instance ID: i-02202675e711cd943
PS E:\CODING\Devops-repo\boto3>
```

You can observe a new instance has been initiated .



- Instead of hardening the credentials in python code locally we can use secrete manager to fetch the secrets from the aws console and include on the top of the python automate code.
- If a amazon resources , ex- ec2 instance try to automatically create a new instance then the instance should have AmazonEc2FullAccess role attached to it .

Task Over